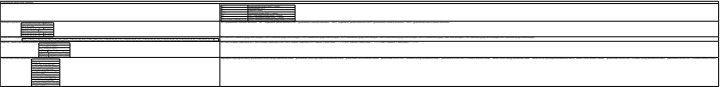


Tables

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Identifier	Page	Conf.	LaTeX source	Rendered	Image
1510.06699_TAB_001	55	—	<pre> \begin{tabular}[t]{ l l } \hline \multicolumn{2}{ l }{Gravitational gauge field variables} \\ \hline \begin{tabular}[t]{ l l } \hline (h_{\mu}) & translational gauge field (h-field) \\ \hline (b^a_{\mu}) & determinant of h_a{}^{\mu} \\ \hline (A^{ab}_{\mu}) & inverse h-field \\ \hline (B_{\mu}) & rotational gauge field (A-field) \\ \hline (V_{\mu}) & WGT dilational gauge field (B-field) \\ \hline A^{\dagger ab}_{\mu} \equiv A^{ab}_{\mu} + 2\eta^{c[ab]}_{\mu} h_c{}^{\nu} V_{\nu} & eWGT dilational gauge field (V-field) \\ \hline \end{tabular} & eWGT extended rotational gauge field (A^{\dagger}-field) \\ \hline \end{tabular} Derivative operators D_a \equiv h_a{}^{\mu} D_{\mu} \equiv h_a{}^{\mu} (\partial_{\mu} + \tfrac{1}{2} A^{ab}_{\mu} \Sigma_{ab}) \partial^a_{\mu} \equiv \partial_{\mu} + w B_{\mu} D^a_{\mu} \equiv h_a{}^{\mu} D^a_{\mu} \equiv h_a{}^{\mu} (\partial^a_{\mu} + \tfrac{1}{2} A^{ab}_{\mu} \Sigma_{ab}) \partial^{\dagger}_{\mu} \equiv \partial_{\mu} - w (V_{\mu} + \tfrac{1}{3} T_{\nu}) D^{\dagger}_{\mu} \equiv h_a{}^{\mu} D^{\dagger}_{\mu} \equiv h_a{}^{\mu} (\partial^{\dagger}_{\mu} + \tfrac{1}{2} A^{\dagger ab}_{\mu} \Sigma_{ab}) D^2_{\mu} \equiv h_a{}^{\mu} D^2_{\mu} \equiv h_a{}^{\mu} (\partial_{\mu} + \tfrac{1}{2} A^{\dagger ab}_{\mu} \Sigma_{ab}) Gauge field strengths \mathcal{R}^{ab}_{cd} = \mathcal{R}^{*ab}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{ab}_{\nu]} + A^a{}_{c[\mu} A^{cb}_{\nu]}) \mathcal{R}^{\dagger ab}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{\dagger ab}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}_{\nu]}) T^a{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu} b^a{}_{\nu]} T^*{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D^a_{[\mu} b^a{}_{\nu]} T^{\dagger a}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D^{\dagger}_{[\mu} b^a{}_{\nu]} T^2{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D^2_{[\mu} b^a{}_{\nu]} \mathcal{H}_{ab} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} B_{\nu]} \mathcal{H}^{\dagger}_{ab} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} (V_{\nu} + \tfrac{1}{3} T_{\nu]}) Reduced A-fields and related quantities c^a{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu} b^a{}_{\nu]} c^*{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial^a_{[\mu} b^a{}_{\nu]} c^{\dagger a}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial^{\dagger}_{[\mu} b^a{}_{\nu]} {}^0A^{ab}_{\mu} \equiv \tfrac{1}{2} b^c{}_{\mu} (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab}) {}^0A^{*ab}_{\mu} \equiv \tfrac{1}{2} b^c{}_{\mu} (c^*{}^{ab}{}_c + c^*{}^b{}_c{}^a - c_c^*{}^{ab}) {}^0A^{\dagger ab}_{\mu} \equiv \tfrac{1}{2} b^c{}_{\mu} (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab}) Currents and related quantities \mathcal{L} \equiv h^{-1} L t^a{}_b \equiv t^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_G / \delta h_a{}^{\mu} s_{ab}{}^c \equiv s_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_G / \delta A^{ab}_{\mu} j^a \equiv j^a{} b^{\mu}{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_G / \delta B_{\mu} t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a V^c - s^{ac}{}_c V_b) j^{\dagger a} \equiv j^a - 2s^{ab}{}_b \tau^a{}_b \equiv \tau^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_M / \delta h_a{}^{\mu} \sigma_{ab}{}^c \equiv \sigma_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_M / \delta A^{ab}_{\mu} \zeta^a \equiv \zeta^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_M / \delta B_{\mu} \tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a V^c - \sigma^{ac}{}_c V_b) \zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b </pre>		Gravitational gauge field variables $h_a{}^{\mu}$ translational gauge field (h-field) h determinant of $h_a{}^{\mu}$ $b^a{}_{\mu}$ inverse h-field A^{ab}_{μ} rotational gauge field (A-field) B_{μ} WGT dilational gauge field (B-field) V_{μ} eWGT dilational gauge field (V-field) $A^{\dagger ab}_{\mu} \equiv A^{ab}_{\mu} + 2\eta^{c[ab]}_{\mu} h_c{}^{\nu} V_{\nu}$ eWGT extended rotational gauge field ($A^{\dagger}$-field) Derivative operators $D_a \equiv h_a{}^{\mu} D_{\mu} \equiv h_a{}^{\mu} (\partial_{\mu} + \frac{1}{2} A^{ab}_{\mu} \Sigma_{ab})$ PGT (generalised) covariant derivative^a $\partial^a_{\mu} \equiv \partial_{\mu} + w B_{\mu}$ WGT 'augmented' partial derivative $D^a_{\mu} \equiv h_a{}^{\mu} D^a_{\mu} \equiv h_a{}^{\mu} (\partial^a_{\mu} + \frac{1}{2} A^{ab}_{\mu} \Sigma_{ab})$ WGT (generalised) covariant derivative^a $\partial^{\dagger}_{\mu} \equiv \partial_{\mu} - w (V_{\mu} + \frac{1}{3} T_{\nu})$ eWGT 'augmented' partial derivative $D^{\dagger}_{\mu} \equiv h_a{}^{\mu} D^{\dagger}_{\mu} \equiv h_a{}^{\mu} (\partial^{\dagger}_{\mu} + \frac{1}{2} A^{\dagger ab}_{\mu} \Sigma_{ab})$ eWGT (generalised) covariant derivative^a $D^2_{\mu} \equiv h_a{}^{\mu} D^2_{\mu} \equiv h_a{}^{\mu} (\partial_{\mu} + \frac{1}{2} A^{\dagger ab}_{\mu} \Sigma_{ab})$ eWGT (generalised) semi-covariant derivative^a Gauge field strengths $\mathcal{R}^{ab}_{cd} = \mathcal{R}^{*ab}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{ab}_{\nu]} + A^a{}_{c[\mu} A^{cb}_{\nu]})$ PGT and WGT rotational gauge field strength^{a, b} $\mathcal{R}^{\dagger ab}_{cd} = 2h_a{}^{\mu} h_b{}^{\nu} (\partial_{[\mu} A^{\dagger ab}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}_{\nu]})$ eWGT rotational gauge field strength^{a, b} $T^a{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D_{[\mu} b^a{}_{\nu]}$ PGT translational gauge field strength^b $T^*{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D^a_{[\mu} b^a{}_{\nu]}$ WGT translational gauge field strength^b $T^{\dagger a}{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D^{\dagger}_{[\mu} b^a{}_{\nu]}$ eWGT translational gauge field strength^b $T^2{}_{bc} = 2h_b{}^{\mu} h_c{}^{\nu} D^2_{[\mu} b^a{}_{\nu]}$ eWGT translational gauge 'semi' field strength^{a, b} $\mathcal{H}_{ab} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} B_{\nu]}$ WGT dilational gauge field strength $\mathcal{H}^{\dagger}_{ab} = 2h_a{}^{\mu} h_b{}^{\nu} \partial_{[\mu} (V_{\nu} + \frac{1}{3} T_{\nu]})$ eWGT dilational gauge field strength Reduced A-fields and related quantities $c^a{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial_{[\mu} b^a{}_{\nu]}$ PGT Ricci rotation coefficients $c^*{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial^a_{[\mu} b^a{}_{\nu]}$ WGT Ricci rotation coefficients $c^{\dagger a}{}_{bc} \equiv 2h_b{}^{\mu} h_c{}^{\nu} \partial^{\dagger}_{[\mu} b^a{}_{\nu]}$ eWGT Ricci rotation coefficients ${}^0A^{ab}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ PGT reduced A-field (when $T^a{}_{bc} \equiv 0$) ${}^0A^{*ab}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^*{}^{ab}{}_c + c^*{}^b{}_c{}^a - c_c^*{}^{ab})$ WGT reduced A-field (when $T^*{}^a{}_{bc} \equiv 0$) ${}^0A^{\dagger ab}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$ eWGT reduced $A^{\dagger}$-field (when $T^{\dagger a}{}_{bc} \equiv 0$) Currents and related quantities $\mathcal{L} \equiv h^{-1} L$ Lagrangian $t^a{}_b \equiv t^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_G / \delta h_a{}^{\mu}$ Lagrangian density $s_{ab}{}^c \equiv s_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_G / \delta A^{ab}_{\mu}$ gravitational sector energy-momentum tensor^c $j^a \equiv j^a{} b^{\mu}{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_G / \delta B_{\mu}$ gravitational sector spin-angular-momentum tensor^c $t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a V^c - s^{ac}{}_c V_b)$ gravitational sector dilation current^c $j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$ gravitational sector eWGT covariant energy-momentum tensor^c $\tau^a{}_b \equiv \tau^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_M / \delta h_a{}^{\mu}$ gravitational sector eWGT covariant dilation current^c $\sigma_{ab}{}^c \equiv \sigma_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_M / \delta A^{ab}_{\mu}$ matter sector energy-momentum tensor^c $\zeta^a \equiv \zeta^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_M / \delta B_{\mu}$ matter sector spin-angular-momentum tensor^c $\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a V^c - \sigma^{ac}{}_c V_b)$ matter sector dilation current^c $\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$ gravitational sector eWGT covariant energy-momentum tensor^c gravitational sector eWGT covariant dilation current^c

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1510.06699_TAB_004	55	—	$\begin{tabular}{t l} \hline \mathcal{R}^{\mathfrak{a}b}_{cd} = \mathcal{R}^{*ab}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}_{\nu]} + A^a{}_{c[\mu} A^{cb}_{\nu]}) \\ \mathcal{R}^{\dagger ab}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}_{\nu]}) \\ \mathcal{T}^a_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a_{\nu]} \\ \mathcal{T}^{*a}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu} b^a_{\nu]} \\ \mathcal{T}^{\dagger a}_{bc} = 2h_b{}^\mu h_c{}^\nu D^{\dagger}_{[\mu} b^a_{\nu]} \\ \mathcal{T}^{\mathfrak{a}}_{bc} = 2h_b{}^\mu h_c{}^\nu D^{\mathfrak{a}}_{[\mu} b^a_{\nu]} \\ \mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]} \\ \mathcal{H}^{\dagger}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3} T_\nu)] \\ \hline \end{tabular}$	$\mathcal{R}^{\mathfrak{a}b}_{cd} = \mathcal{R}^{*ab}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}_{\nu]} + A^a{}_{c[\mu} A^{cb}_{\nu]})$	$\mathcal{R}^{ab}_{cd} = \mathcal{R}^{*ab}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{ab}_{\nu]} + A^a{}_{c[\mu} A^{cb}_{\nu]})$ $\mathcal{R}^{\dagger ab}_{cd} = 2h_a{}^\mu h_b{}^\nu (\partial_{[\mu} A^{\dagger ab}_{\nu]} + A^{\dagger a}{}_{c[\mu} A^{\dagger cb}_{\nu]})$ $\mathcal{T}^a_{bc} = 2h_b{}^\mu h_c{}^\nu D_{[\mu} b^a_{\nu]}$ $\mathcal{T}^{*a}_{bc} = 2h_b{}^\mu h_c{}^\nu D^*_{[\mu} b^a_{\nu]}$ $\mathcal{T}^{\dagger a}_{bc} = 2h_b{}^\mu h_c{}^\nu D^{\dagger}_{[\mu} b^a_{\nu]}$ $\mathcal{T}^{\mathfrak{a}}_{bc} = 2h_b{}^\mu h_c{}^\nu D^{\mathfrak{a}}_{[\mu} b^a_{\nu]}$ $\mathcal{H}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} B_{\nu]}$ $\mathcal{H}^{\dagger}_{ab} = 2h_a{}^\mu h_b{}^\nu \partial_{[\mu} (V_\nu + \frac{1}{3} T_\nu)]$
1510.06699_TAB_005	55	—	$\begin{tabular}{t l} \hline h_{\{b\}\}^{\{\mu\}} h_{\{c\}\}^{\{\nu\}} \partial_{[\mu} b^a_{\nu]} \equiv 2 h_{\{b\}\}^{\{\mu\}} h_{\{c\}\}^{\{\nu\}} \partial_{[\mu} b^a_{\nu]} \\ c^{*a}_{bc} \equiv 2 h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a_{\nu]} \\ c^{\dagger a}_{bc} \equiv 2 h_b{}^\mu h_c{}^\nu \partial_{[\mu}^{\dagger} b^a_{\nu]} \\ {}^0 A^{ab}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab}) \\ {}^0 A^{*ab}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c^{*ab}) \\ {}^0 A^{\dagger ab}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab}) \\ \hline \end{tabular}$	$c^a_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a_{\nu]}$ $c^{*a}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a_{\nu]}$ $c^{\dagger a}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^{\dagger} b^a_{\nu]}$ ${}^0 A^{ab}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ ${}^0 A^{*ab}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c^{*ab})$ ${}^0 A^{\dagger ab}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$	$c^a_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu} b^a_{\nu]}$ $c^{*a}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^* b^a_{\nu]}$ $c^{\dagger a}_{bc} \equiv 2h_b{}^\mu h_c{}^\nu \partial_{[\mu}^{\dagger} b^a_{\nu]}$ ${}^0 A^{ab}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{ab}{}_c + c^b{}_c{}^a - c_c{}^{ab})$ ${}^0 A^{*ab}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{*ab}{}_c + c^{*b}{}_c{}^a - c_c^{*ab})$ ${}^0 A^{\dagger ab}_{\mu} \equiv \frac{1}{2} b^c{}_{\mu} (c^{\dagger ab}{}_c + c^{\dagger b}{}_c{}^a - c_c^{\dagger ab})$

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1510.06699_TAB_006	55	—	$\begin{tabular}[t]{ l } \hline \mathcal{L} \equiv h^{-1} L \\ \equiv t^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_G / \delta h_a{}^{\mu} \\ s_{ab}{}^c \equiv s_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_G / \delta A^{ab}{}_{\mu} \\ j^a \equiv j^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_G / \delta B_{\mu} \\ t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b) \\ j^{\dagger a} \equiv j^a - 2s^{ab}{}_b \\ \tau^a{}_b \equiv \tau^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_M / \delta h_a{}^{\mu} \\ \sigma_{ab}{}^c \equiv \sigma_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_M / \delta A^{ab}{}_{\mu} \\ \zeta^a \equiv \zeta^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_M / \delta B_{\mu} \\ \tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b) \\ \zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b \end{tabular}$	<table><tr><td>L</td></tr><tr><td>$\mathcal{L} \equiv h^{-1} L$</td></tr><tr><td>$t^a{}_b \equiv t^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_G / \delta h_a{}^{\mu}$</td></tr><tr><td>$s_{ab}{}^c \equiv s_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_G / \delta A^{ab}{}_{\mu}$</td></tr><tr><td>$j^a \equiv j^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_G / \delta B_{\mu}$</td></tr><tr><td>$t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$</td></tr><tr><td>$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$</td></tr><tr><td>$\tau^a{}_b \equiv \tau^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_M / \delta h_a{}^{\mu}$</td></tr><tr><td>$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_M / \delta A^{ab}{}_{\mu}$</td></tr><tr><td>$\zeta^a \equiv \zeta^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_M / \delta B_{\mu}$</td></tr><tr><td>$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$</td></tr><tr><td>$\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$</td></tr></table>	L	$\mathcal{L} \equiv h^{-1} L$	$t^a{}_b \equiv t^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_G / \delta h_a{}^{\mu}$	$s_{ab}{}^c \equiv s_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_G / \delta A^{ab}{}_{\mu}$	$j^a \equiv j^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_G / \delta B_{\mu}$	$t^{\dagger a}{}_b \equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b)$	$j^{\dagger a} \equiv j^a - 2s^{ab}{}_b$	$\tau^a{}_b \equiv \tau^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_M / \delta h_a{}^{\mu}$	$\sigma_{ab}{}^c \equiv \sigma_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_M / \delta A^{ab}{}_{\mu}$	$\zeta^a \equiv \zeta^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_M / \delta B_{\mu}$	$\tau^{\dagger a}{}_b \equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b)$	$\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$	$\begin{aligned} L \\ \mathcal{L} &\equiv h^{-1} L \\ t^a{}_b &\equiv t^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_G / \delta h_a{}^{\mu} \\ s_{ab}{}^c &\equiv s_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_G / \delta A^{ab}{}_{\mu} \\ j^a &\equiv j^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_G / \delta B_{\mu} \\ t^{\dagger a}{}_b &\equiv t^a{}_b + 2(s_{cb}{}^a \mathcal{V}^c - s^{ac}{}_c \mathcal{V}_b) \\ j^{\dagger a} &\equiv j^a - 2s^{ab}{}_b \\ \tau^a{}_b &\equiv \tau^a{}_{\mu} h_b{}^{\mu} \equiv h_b{}^{\mu} \delta \mathcal{L}_M / \delta h_a{}^{\mu} \\ \sigma_{ab}{}^c &\equiv \sigma_{ab}{}^{\mu} b^c{}_{\mu} \equiv b^c{}_{\mu} \delta \mathcal{L}_M / \delta A^{ab}{}_{\mu} \\ \zeta^a &\equiv \zeta^{\mu} b^a{}_{\mu} \equiv b^a{}_{\mu} \delta \mathcal{L}_M / \delta B_{\mu} \\ \tau^{\dagger a}{}_b &\equiv \tau^a{}_b + 2(\sigma_{cb}{}^a \mathcal{V}^c - \sigma^{ac}{}_c \mathcal{V}_b) \\ \zeta^{\dagger a} &\equiv \zeta^a - 2s^{ab}{}_b \end{aligned}$
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$\zeta^{\dagger a} \equiv \zeta^a - 2s^{ab}{}_b$																	

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1510.06699_TAB_007	—	—	\begin{tabular}{ll} {\em Gravitational gauge field variables} & \\\ \${h_a}^{\mu}\$ & translational gauge field (\$h\$-field) \\\ \$h\$ & determinant of \${h_a}^{\mu}\$ \\\ \${b^a}_{\mu}\$ & inverse \$h\$-field \\\ \${A^{\dagger}ab}_{\mu}\$ & rotational gauge field (\$A\$-field) \\\ \$B_{\mu}\$ & WGT dilational gauge field (\$B\$-field) \\\ \$V_{\mu}\$ & eWGT dilational gauge field (\$V\$-field) \\\ \${A^{\dagger}ab}_{\mu} \equiv {A^{\dagger}ab}_{\mu} + 2\eta^{ca}{b^b}_{\mu}\$ & \\\ \$\nu V_{\nu}\$ & eWGT extended rotational gauge field (\$A^{\dagger}\$-field) \\\ & \\\ {\em Derivative operators} & \\\ \$\mathcal{D}_a \equiv {h_a}^{\mu} D_{\mu} \equiv {h_a}^{\mu} (\partial_{\mu} + \frac{1}{2}{A^{\dagger}ab}_{\mu}\Sigma_{ab})\$ & \\\ PGT (generalised) covariant derivative\footnotemark[1] \\\ \$\partial^{\ast}_{\mu} \equiv \partial_{\mu} + wB_{\mu}\$ & \\\ WGT 'augmented' partial derivative \\\ \$\mathcal{D}^{\ast}_a \equiv {h_a}^{\mu} D^{\ast}_{\mu} \equiv {h_a}^{\mu} (\partial^{\ast}_{\mu} + \frac{1}{2}{A^{\dagger}ab}_{\mu}\Sigma_{ab})\$ & \\\ WGT (generalised) covariant derivative\footnotemark[1] \\\ \$\partial^{\dagger}_{\mu} \equiv \partial_{\mu} - w(V_{\mu} + \frac{1}{3}T_{\mu})\$ & \\\ eWGT 'augmented' partial derivative \\\ \$\mathcal{D}^{\dagger}_a \equiv {h_a}^{\mu} D^{\dagger}_{\mu} \equiv {h_a}^{\mu} (\partial^{\dagger}_{\mu} + \frac{1}{2}{A^{\dagger}ab}_{\mu}\Sigma_{ab})\$ & \\\ eWGT (generalised) covariant derivative\footnotemark[1] \\\ \$\mathcal{D}^{\natural}_a \equiv {h_a}^{\mu} D^{\natural}_{\mu} \equiv {h_a}^{\mu} (\partial_{\mu} + \frac{1}{2}{A^{\dagger}ab}_{\mu}\Sigma_{ab})\$ & \\\ eWGT (generalised) semi-covariant derivative\footnotemark[1] \\\ & \\\ {\em Gauge field strengths} & \\\ \$\mathcal{R}^{\dagger}_{ab} = \mathcal{R}^{\dagger}_{ab} - \partial^{\ast}ab = 2{h_a}^{\mu}{h_b}^{\nu}(\partial_{\mu}{b^a}_{\nu}) + {A^a}_{c\mu}{A^{\dagger}cb}_{\nu})\$ & \\\ PGT and WGT rotational gauge field strength\footnotemark[1], \\\ \footnotemark[2] \\\ \$\mathcal{R}^{\dagger\dagger}_{ab} = 2{h_a}^{\mu}{h_b}^{\nu}(\partial_{\mu}{A^{\dagger}ab}_{\nu}) + {A^{\dagger}a}_{c\mu}{A^{\dagger}daggercb}_{\nu})\$ & \\\ eWGT rotational gauge field strength\footnotemark[1], \\\ \footnotemark[2] \\\ \$\mathcal{T}^a_{bc} = 2{h_b}^{\mu}{h_c}^{\nu}D_{\mu}{b^a}_{\nu})\$ & \\\ PGT translational gauge field strength\footnotemark[2] \\\ \$\mathcal{T}^{\dagger}_{bc} = 2{h_b}^{\mu}{h_c}^{\nu}D^{\ast}_{\mu}{b^a}_{\nu})\$ & \\\ WGT translational gauge field strength \\\ \footnotemark[2] \\\ \$\mathcal{T}^{\dagger\dagger}_{bc} = 2{h_b}^{\mu}{h_c}^{\nu}D^{\dagger}_{\mu}{b^a}_{\nu})\$ & \\\ eWGT translational gauge field strength\footnotemark[2] \\\ \\\ \$\mathcal{T}^{\natural}_a = 2{h_b}^{\mu}{h_c}^{\nu}D^{\natural}_{\mu}{b^a}_{\nu})\$ & \\\ eWGT translational gauge 'semi' field strength\footnotemark[1],\footnotemark[2] \\\ \\\ \$\mathcal{H}_{ab} = 2{h_a}^{\mu}{h_b}^{\nu}\partial_{\mu}{B_{\nu}}\$ & \\\ WGT dilational gauge field strength \\\ \$\mathcal{H}^{\dagger}_{ab} = 2{h_a}^{\mu}{h_b}^{\nu}\partial_{\mu}(V_{\nu} + \frac{1}{3}T_{\nu})\$ & \\\ eWGT dilational gauge field strength \\\ & \\\ {\em Reduced \$A\$-fields and related quantities} & \\\ \${c^a}_{bc} \equiv 2{h_b}^{\mu}{h_c}^{\nu}\partial_{\mu}{b^a}_{\nu})\$ & \\\ PGT Ricci rotation coefficients \\\ \${c^{\ast}a}_{bc} \equiv 2{h_b}^{\mu}{h_c}^{\nu}\partial^{\ast}_{\mu}{b^a}_{\nu})\$ & \\\ WGT Ricci rotation coefficients \\\ \${c^{\dagger}a}_{bc} \equiv 2{h_b}^{\mu}{h_c}^{\nu}\partial^{\dagger}_{\mu}{b^a}_{\nu})\$ & \\\ eWGT Ricci rotation coefficients \\\ \${c^{\dagger\dagger}a}_{bc} \equiv 2{h_b}^{\mu}{h_c}^{\nu}\partial^{\dagger\dagger}_{\mu}{b^a}_{\nu})\$ & \\\ eWGT Ricci rotation coefficients \\\ \$\{^0A^{\dagger}ab\}_{\mu} \equiv \frac{1}{2}{b^c}_{\mu}({c^{\dagger}ab}_c + {c^{\dagger}b}_c{}^a - {c^{\dagger}ab}_c)\$ & \\\ PGT reduced \$A\$-field (when \$\mathcal{T}^a_{bc} \equiv 0\$) \\\ \$\{^0A^{\dagger}ab\}_{\mu} \equiv \frac{1}{2}{b^c}_{\mu}({c^{\ast}ab}_c + {c^{\ast}b}_c{}^a - {c^{\ast}ab}_c)\$ & \\\ WGT reduced \$A\$-field (when \$\mathcal{T}^{\ast}_{bc} \equiv 0\$) \\\ & \\\ \$\{^0A^{\dagger\dagger}ab\}_{\mu} \equiv \frac{1}{2}{b^c}_{\mu}\$ \end{tabular}	(not rendered)	—

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